

IPv4 ADDRESSING

1. Convert the following IP address to its 32-bit binary representation: 172.16.5.10.
Write out each octet in binary, separated by dots.

-> To convert **172.16.5.10** into 32-bit binary, convert each octet separately:

- **172** → 10101100
- **16** → 00010000
- **5** → 00000101
- **10** → 00001010

Answer

172.16.5.10

↓

10101100.00010000.00000101.00001010

So, the **32-bit binary representation** is:

10101100.00010000.00000101.00001010

2. Given the binary IP address 11000000.10101000.00000001.00001111, convert it back to standard dotted decimal notation.

-> Convert each 8-bit binary octet to decimal:

- 11000000 → **192**
- 10101000 → **168**
- 00000001 → **1**
- 00001111 → **15**

Answer

11000000.10101000.00000001.00001111

= 192.168.1.15 

3. You are given a list of 10 IP addresses used by different apps (e.g., 192.168.43.1, 8.8.8.8, 10.0.5.25, 172.20.14.2, 203.0.113.5, 172.15.255.255, 192.0.2.10, 192.168.1.100, 123.45.67.89, 172.16.0.1). For each, identify if it is a private or public IP address and explain your reasoning.

-> To classify IPv4 addresses, the **private address ranges** are:

- **10.0.0.0 – 10.255.255.255** (10.0.0.0/8)
- **172.16.0.0 – 172.31.255.255** (172.16.0.0/12)
- **192.168.0.0 – 192.168.255.255** (192.168.0.0/16)

Any address outside these ranges is generally **public**, although some addresses are reserved for documentation/testing rather than actually routable on the public Internet.

IP Address	Classification	Reason
192.168.43.1	 Private	Falls within 192.168.0.0/16
8.8.8.8	 Public	Not within any private range; it is Google's public DNS address
10.0.5.25	 Private	Falls within 10.0.0.0/8
172.20.14.2	 Private	Falls within 172.16.0.0/12
203.0.113.5	 Reserved	203.0.113.0/24 is reserved for documentation (TEST-NET-3)
172.15.255.255	 Public	172.15.x.x is outside the private 172.16–172.31 range
192.0.2.10	 Reserved	192.0.2.0/24 is reserved for documentation (TEST-NET-1)
192.168.1.100	 Private	Falls within 192.168.0.0/16
123.45.67.89	 Public	Not within any private range
172.16.0.1	 Private	Falls within 172.16.0.0/12

Final classification

Private:

192.168.43.1, 10.0.5.25, 172.20.14.2, 192.168.1.100, 172.16.0.1

Public:

8.8.8.8, 172.15.255.255, 123.45.67.89

Reserved for documentation:

203.0.113.5, 192.0.2.10

Important: 172.15.255.255 is a useful trick question. Only **172.16.0.0 through 172.31.255.255** are private—not the entire 172.x.x.x range.

4. For the CIDR notations /20, /25, and /30, calculate the total number of usable IP addresses in each subnet. Show your working for each calculation.

Hint: Usable addresses = $(2^{\text{number of host bits}}) - 2$ (for network and broadcast addresses).

-> For IPv4, there are **32 total bits**. The CIDR number tells us how many bits are used for the network, so:

Host bits = 32 – CIDR prefix

Usable IP addresses = $2^{\text{(host bits)}} - 2$

1. /20

- Host bits = $32 - 20 = 12$
- Total addresses = $2^{12} = 4096$
- Usable addresses = $4096 - 2$

✓ **Usable IP addresses = 4,094**

2. /25

- Host bits = $32 - 25 = 7$
- Total addresses = $2^7 = 128$
- Usable addresses = $128 - 2$

✓ **Usable IP addresses = 126**

3. /30

- Host bits = $32 - 30 = 2$
- Total addresses = $2^2 = 4$
- Usable addresses = $4 - 2$

✓ **Usable IP addresses = 2**

Final Answer

CIDR	Host Bits	Total Addresses	Usable Addresses
/20	12	4,096	4,094

CIDR	Host Bits	Total Addresses	Usable Addresses
/25	7	128	126
/30	2	4	2

The **2 addresses excluded** in each traditional IPv4 subnet are the **network address** and **broadcast address**.

- Imagine you are setting up a small LAN for a gaming night with friends using your WiFi router at home. Assign static private IP addresses to 5 devices (laptop, phone, tablet, smart TV, and PlayStation) using the 192.168.0.0/24 range. List the IP you would assign to each device and explain why you chose that range.

-> Static IP Address Plan for a Gaming-Night LAN

For the five devices, I would use the **192.168.0.0/24** private network and assign addresses like this:

Device	Static IP Address	Subnet Mask
 Laptop	192.168.0.10	255.255.255.0
 Phone	192.168.0.11	255.255.255.0
 Tablet	192.168.0.12	255.255.255.0
 Smart TV	192.168.0.13	255.255.255.0
 PlayStation	192.168.0.14	255.255.255.0

Why choose 192.168.0.0/24?

- Private range:** 192.168.0.0/24 is part of the IPv4 private address space, so these addresses are intended for local networks and aren't directly routable on the public Internet.
- Enough addresses:** /24 provides **256 total addresses** (192.168.0.0–192.168.0.255), with **254 traditionally usable host addresses**.
- Easy organization:** Using .10–.14 makes the five gaming-night devices easy to remember and manage.
- Same LAN:** All five devices are in the same subnet, allowing them to communicate directly with each other for LAN gaming and media sharing.

